Reg. No. | | | | | | | | | | |

[13EC7608/12EC7608]

SEVENTH SEMESTER B.Tech/B.E. UNIVERSITY EXAMINATION, DECEMBER - 2016

IMAGE PROCESSING

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	a.	Describe fundamental steps in Digital Image Processing.	07
	b.	What is Weber Ratio? Explain how range of subjective brightness sensations showing a particular adaption level?	07
			07
2.	a.	Describe a Simple Image Formation Model. Explain Adjacency, Connectivity, Regions and Boundaries.	07
	b.	Describe Single Imaging Sensor, Line Sensor and Array Sensor.	07
3.	a.	Explain 2D orthogonal and unitary transform.	07
	b.	With necessary equations explain discrete cosine transform.	07
4.	a.	Describe Basic Intensity Transformation Functions.	07
	b.	Explain Histogram Equalization.	07
5.	a.	Describe Histogram Matching.	07
	b.	Explain the process of Smoothing Linear Filters.	07
6.	a.	Describe a Model of the Image Degradation / Restoration Process. Also explain PDF of Rayleigh Noise.	07
	b.	Describe Different Mean Filters and Discuss its effects.	07
7.	a.	Describe Color Fundamentals with trichromatic coefficients.	07
	b.	Explain basics of full color image processing. Also explain CMY and CMYK color models.	07
8.	a.	Discuss fidelity criteria and Shannon's first theorem.	07
	b.	Describe functional block diagram of a general image compression system.	07